

jects large enough with correct temperature information."

Some physiological activities, particularly responses such as fever in human beings and other warm-blooded animals, can also be monitored with thermographic imaging. Cooled IR cameras can be found at major astronomy research telescopes, even those that are not IR telescopes.

Commercial applications. "Due to high cost and sophistication of thermal imagers, civil market is presently very limited," says Prijo Samuel, senior consultant global electronics 6Wresearch. But even with limited numbers, there is demand for such equipment. These equipment are highly precise in their measurements. "Products being developed for electrical and mechanical applications can measure temperature as per specified range with an accuracy of $\pm 1.5^{\circ}\text{C}$," adds Singh.

"Whenever used, commercial applications require long operating hours. Micro-bolometers are good for that since no maintenance is required," adds Israely.

Redirecting traffic. Heavy traffic is now being controlled via thermal imagers as well. Manufacturers have come up with equipment specifically designed to cover rush-hour

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traffic in urban environments. These provide very accurate and real-time solutions for traffic controllers to dynamically reroute traffic.

FLIR TraqOne is one such thermal imager module. It connects to a city's traffic signal controller to allow for a more dynamic control of traffic signals based on traffic and pedestrian patterns and intensity. Using traffic flow information collected through a Wi-Fi tracking technology, it can also measure traffic delay times at intersections.

Defence involvement. Designing defence equipment requires a great deal of innovation and technological know-how. As Israely says, "Defence-related projects are often related to customising existing solutions to specific needs."

"Thermal imaging systems such as thermal imaging fire control and thermal imaging sights were

deployed in main battle tanks and mechanised infantry vehicles in recent years," Rajat Kharbanda, senior consultant, research and consulting, 6Wresearch, adds.

Israely adds, "Driver's vision enhancement kits are built to withstand the harsh conditions that involve armoured vehicle movement through the battlefield."

Thermal imaging modules, and not camera core ones, are being called in for the defence sector. This allows for ease of use for defence personal. "MEOS modular EO/IR system is proven for border and coastal security as well as perimeter protection of critical sites including military bases, offshore oil/gas rigs and more," mentions CONTROP's vice president of marketing, Johnny Carni, in a press release.

The future looks hot

Advances like easy-to-use accessories for smartphones have further increased the ease of use and made the devices available to the user. With applications ranging from security applications to sports referrals, newer uses are being considered daily.

"There is growing demand for handheld thermal imaging devices for industrial applications, and for cooled thermal imaging systems for fighter aircraft and armoured divisions," adds Samuel. Requirement for portable equipment is on the rise, and thermal imagers will definitely not be left behind. **EFY**

Parameters defining a thermal camera

Due to heavy surveillance usage, several parameters have to be defined in order to consider the best possible equipment for use.

Frame rate. An important parameter for a thermal imager is frame rate. Today, we employ imagers with frame rates from 9Hz to 60Hz. Where a 9Hz video would look like a series of fast-moving images, a 60Hz video looks like a regular video with different colours.

Sensor. Since heat measurements are based on the sensor, the entire device could be designed around it. This would also define the size of the end camera, which would become another factor in application.

Pixel size. "We see a constant decrease in pixel size of micro-bolometers today," says Israely. This reduction in pixel size leads to a higher number of pixels in the same area. This leads to further image clarity.

Coverage area. This covers the image quality while considering a thermal image. A high pixel size would provide a wider FOV along with a larger coverage area in a single camera.

Power consumption. This is a matter of consideration with every device. It essentially controls the usage of the device. A low-power usage device is used in portable cameras, whereas others find applications with dedicated power supplies.